CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in agentic systems and their evaluation, from multi-agent LLM pipelines and context curation to ground-truth evaluation and deployment behind APIs and containers. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by research-adjacent work on RAG evaluation with PyTorch and Hugging Face, and strong Python development.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Designed and optimized a multi-agent system (LangGraph: entity-extraction, relationship-analysis, consolidation and reflection agents in parallel) for multi-step knowledge extraction from complex regulatory and technical documents, driving precision/recall from 33%/45% to 93%/90% against an SAP-system-derived ground truth.
- Sole developer owning the full path from raw documents to production – data ingestion, chunking, context curation through prompt engineering, and indexing (agentic, deterministic, monolithic strategies) – reporting directly to the CTO and working with SAP domain experts.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) query service, a developer-facing abstraction exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, letting domain experts validate and approve knowledge-graph changes.
- Delivered as an AI-augmented engineer – using Claude Code, custom skills and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, RAG, GraphRAG, Vector Search, Neo4j, MongoDB, Postgres, FastAPI, Docker, Kubernetes, Azure, React**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed and deployed an agentic AI system with LangGraph that routes each request across vector database queries, general inference and summarization, after working with business stakeholders to map their decision-making workflows.
- Developed a React-based front-end and a Python backend API with FastAPI, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS cloud infrastructure using CDK, following clean code practices. Version-controlled via GitLab, with CI/CD integration.
- **Tech stack: Python(Backend), LangGraph, LangChain, FastAPI, Docker, AWS, CDK, EC2, Lambda, GitLab CI/CD, Postman, React(Frontend)**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated evaluation of RAG generation and retrieval with RAGAS and PytrecEval, scoring model answers against chemist-provided ground truth to locate failures in retrieval versus generation.
- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.

- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.
- **Tech stack: Python, PyTorch, Hugging Face Transformers, Scikit-Learn, TensorFlow, LangChain, RAGAS, PytrecEval, AWS (S3, EC2, VPC, Lambda, SQS, DynamoDB, CDK), Azure AI Search, Docker, Streamlit, Vector Databases**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Agent Eval Harness | Python | PyTorch | LangGraph | Weights & Biases

- Agentic harness around a small PyTorch time-series model exposed as a tool: an LLM planner splits long-horizon questions into hierarchical sub-queries and curates examples per call for in-context learning; release criteria as code (golden tasks, pass rates over repeated runs, failure-mode tags) logged to Weights & Biases and gated in GitHub Actions.

TECHNICAL SKILLSET

- **Agentic AI & LLMs:** Agentic systems (LangGraph, multi-agent), LangChain, LiteLLM, MCP, RAG, GraphRAG, prompt engineering, evaluation (RAGAS, PytrecEval), Hugging Face Transformers, QLoRA fine-tuning
- **ML & Python:** Python, PyTorch, TensorFlow/Keras, Scikit-Learn, XGBoost, time series (ARIMA, Prophet, LSTM)
- **Serving, Cloud & Data:** FastAPI, Python microservices, Docker, Kubernetes, AWS (S3, EC2, Lambda, SQS, DynamoDB, CDK), Azure, Git, GitLab CI/CD, SQL, Postgres, MongoDB, Neo4j, Vector Databases (Qdrant, Pinecone, Azure AI Search), Agile
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

B.Tech Mechanical Engineering (Minor in Computer Science & Engineering) — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India